7.1 Structure of ecosystem

Ecosystem comprises of two major components. They are:

i) **Abiotic (non-living) components**: It includes climatic factors (air, water, sunlight, rainfall, temperature and humidity), edaphic factors (soil air, soil water and pH of soil),topography (latitude, altitude), organic components (carbohydrates, proteins, lipids and humic substances) and inorganic substances (C, H, O, N and P). Abiotic components play vital role in any ecosystem and hence the total inorganic substances present in any ecosystem at a given time is called **standing quality** (or) **standing state**.

ii) **Biotic (living) components**: It **i**ncludes all living organisms like plants, animals, fungi and bacteria. They form the trophic structures of any ecosystem. On the basis of nutritional relationships, trophic levels of an ecosystem have two components. (1) autotrophic components and (2) heterotrophic components.

(1) Autotrophic components: Autotrophs are organisms which can manufacture the organic compounds from simple inorganic components through a process called photosynthesis. In most of the ecosystems, green plants are the autotrophs and are also called **producers**.

(2) Heterotrophic components: Those organisms which consume the producers are called **consumers** and can be recognized into macro and micro consumers. **Macroconsumers** refer to herbivores, carnivores and omnivores (primary, secondary and tertiary consumers). **Microconsumers** are called decomposers. Decomposers are organisms that decompose the dead plants and animals to release organic and inorganic nutrients into the environment which are again reused by plants. Example: Bacteria, Actinomycetes and Fungi.

The amount of living materials present in a population at any given time is known as **standing crop**, which may be expressed in terms of number or biomass per unit area. **Biomass** can be measured as fresh weight or dry weight or carbon weight of organisms. Biotic components are essential to construct the food chain, food web and ecological pyramids.

7.2 Functions of ecosystem

The function of ecosystem include creation of energy creation, sharing of energy and cycling of materials between the living and nonliving components of an ecosystem.

Before studying the productivity in any ecosystem, we should understand the essential role of sunlight used by producers of the first trophic level. The quantity of sunlight is directly proportional to the production of energy by plants.

7.2.1 Photosynthetically Active Radiation (PAR)

The amount of light available for photosynthesis of plants is called **P**hotosynthetically **A**ctive **R**adiation (PAR) which is from of 400-700 nm in wave length. It is essential for photosynthesis and plant growth. PAR is not always constant because of clouds, tree shades, air, dust particles, seasons, latitudes and length of the daylight availability. Generally plants absorb more blue and red light for efficient photosynthesis.

Of the total sunlight, 34 percent that reaches the atmosphere is reflected back into the atmosphere, moreover 10% is held by ozone, water vapours and atmospheric gases and the remaining 56% reaches the earth's surface. Out of this 56%, only 2 – 10% of the solar energy is used by green plants for photosynthesis while the remaining portion is dissipated as heat.

PAR is generally expressed in millimoles / square meter / second by using silicon photo voltic detectors which detect only 400 – 700 nm wavelength of light. PAR values range from 0 to 3000 millimoles /square meter / second.

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